

STA322 - Project 1

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Introduction

Our project uses a two-stage probability proportional to size (PPS) cluster sample to estimate characteristics of faculty in Trinity College of Arts & Sciences at Duke University. We aimed to estimate four characteristics of this population: the fraction of faculty who are women, the fraction with at least one grant ongoing in 2026, the average number of years since receiving a Ph.D. among faculty who hold one, and the average number of years since last publication among faculty who have published. To answer these questions, we aimed for a representative sample of about 100 faculty, drawn in a way that limited how many departmental websites we had to visit.

Sampling design

Our goal was a final sample of 100 faculty, which we obtained in two stages. In the first stage, we selected 20 of the 41 Trinity departments using systematic PPS sampling without replacement, with each department's number of faculty as its measure of size. In the second stage, we selected 5 faculty from each chosen department by simple random sampling without replacement. This gives a final sample of $20 \times 5 = 100$ faculty.

```
library(survey)
```

```
Loading required package: grid
```

```
Loading required package: Matrix
```

```
Loading required package: survival
```

```
Attaching package: 'survey'
```

The following object is masked from 'package:graphics':

dotchart

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.1      v tibble     3.2.1
v lubridate  1.9.3      v tidyr      1.3.1
v purrr      1.0.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
x tidyr::expand() masks Matrix::expand()
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
x tidyr::pack()    masks Matrix::pack()
x tidyr::unpack() masks Matrix::unpack()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(pps)
library(dplyr)
```

```
# load and reorganize data
```

```
trinity <- read_csv("trinity_faculty.csv")
```

```
Rows: 41 Columns: 5
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Dept name
```

```
dbl (4): Stratum #, # of faculty, Proportional allocation, Rounded allocation
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
trinity <- trinity |>
  select(department = "Dept name", M_i = "# of faculty")
```

```

# certainty departments (pi_i = 1), same 7 as in the sampling step
certainty_depts <- c("Economics", "Physics", "Romance Studies", "Music",
                    "Biology", "Mathematics", "Psychology & Neuroscience")

certainty <- trinity |>
  filter(department %in% certainty_depts) |>
  mutate(pi_i = 1)

# remaining departments: 13 more selected, pi_i = 13 * M_i / sum(M_i)
remaining <- trinity |>
  filter(!department %in% certainty_depts)
remaining$pi_i <- 13 * remaining$M_i / sum(remaining$M_i)

dept_info <- bind_rows(certainty, remaining)

n_dept <- 20
t_x <- sum(trinity$M_i)
trinity$pi_i <- n_dept * trinity$M_i / t_x

max(trinity$pi_i) # should be 1.378, so some departments exceed 1

```

```
[1] 1.378151
```

Let M_i be the number of eligible faculty in department i , and let $M_0 = \sum_{i=1}^{41} M_i$ be the total number of eligible faculty across all 41 departments. If no department were large enough to require certainty selection, the first-order inclusion probability for department i under PPS sampling would be

$$\pi_i = \frac{20}{M_0} M_i$$

For six of the largest departments, this formula gave $\pi_i > 1$, which is not a valid probability.

```

# dealing with certainty cases

# separate initial certainty and non-certainty departments
certainty <- trinity |>
  filter(pi_i >= 1) |>
  mutate(pi_i = 1)

remaining <- trinity |>
  filter(pi_i < 1)

```

```

n_remaining <- n_dept - nrow(certainty) # calculate number of additional departments to select
remaining$pi_i <- n_remaining * remaining$M_i / sum(remaining$M_i) # recalculate pi_i for remaining
# if recalculation creates additional certainty PSUs, repeat until all remaining probabilities are < 1
while (any(remaining$pi_i >= 1)) {
  new_certainty <- remaining |>
    filter(pi_i >= 1) |>
    mutate(pi_i = 1)
  certainty <- bind_rows(certainty, new_certainty)
  remaining <- remaining |>
    filter(pi_i < 1)
  n_remaining <- n_dept - nrow(certainty)
  remaining$pi_i <- n_remaining * remaining$M_i / sum(remaining$M_i)
}

# results check
certainty |>
  select(department, M_i, pi_i) # 7 certainty cases identified (1 additional throughout recalculation)

```

```

# A tibble: 7 x 3
  department      M_i pi_i
  <chr>         <dbl> <dbl>
1 Biology         77     1
2 Economics        81     1
3 Mathematics      82     1
4 Music           62     1
5 Physics          60     1
6 Romance Studies  66     1
7 Psychology & Neuroscience 59     1

```

```
nrow(certainty)
```

```
[1] 7
```

```
n_remaining
```

```
[1] 13
```

```
max(remaining$pi_i)
```

```
[1] 0.9246088
```

We treated these departments as certainty PSUs and set $\pi_i = 1$ for them. We then recalculated the inclusion probabilities for the remaining departments, using the number of selections still needed and the total number of faculty in the remaining departments. This pushed one more department above 1, so we repeated the process until every remaining probability was below 1. In the end, 7 departments were certainty PSUs, leaving 13 departments to be selected.

```
set.seed(919)
# select remaining cases with pps & add certainty cases

selected_depts <- ppss(remaining$M_i, n_remaining) # returns numbers of selected rows

pps_remaining <- remaining[selected_depts, ] # keep selected departments

pps_sample <- bind_rows(certainty, pps_remaining) # combine certainty psus with pps-selected

nrow(pps_sample) # check 20 departments were selected
```

```
[1] 20
```

```
# view selected departments
pps_sample |>
  select(department, M_i, pi_i)
```

```
# A tibble: 20 x 3
  department      M_i pi_i
  <chr>      <dbl> <dbl>
1 Biology          77 1
2 Economics         81 1
3 Mathematics       82 1
4 Music            62 1
5 Physics          60 1
6 Romance Studies   66 1
7 Psychology & Neuroscience 59 1
8 Art, Art History & Visual Studies 44 0.814
9 Asian & Middle Eastern Studies 45 0.832
10 Cinematic Arts    7 0.129
```

11 Computer Science	48 0.888
12 Dance Program	21 0.388
13 English	50 0.925
14 Gender, Sexuality & Feminist Studies	13 0.240
15 History	47 0.869
16 Philosophy	24 0.444
17 Political Science	43 0.795
18 Sociology	35 0.647
19 Statistical Science	32 0.592
20 Writing and Rhetoric Program	31 0.573

We set a random seed (919) before selecting the departments so that our sample can be reproduced. [Only if the shuffle is in your code: Before the systematic PPS selection, we also randomly reordered the 41 departments, so the original order of the sampling frame would not influence which departments were chosen.] The 13 non-certainty departments selected this way, together with the 7 certainty departments, gave us our sample of 20 departments.

```
# stage 2 srs

# srs 5 faculty from each department
pps_sample$faculty_numbers <- lapply(
  pps_sample$M_i,
  function(M_i) sample(1:M_i, 5)
)

# view numbers of faculty selected
pps_sample |>
  select(department, M_i, pi_i, faculty_numbers)
```

```
# A tibble: 20 x 4
  department      M_i pi_i faculty_numbers
  <chr>      <dbl> <dbl> <list>
1 Biology      77 1    <int [5]>
2 Economics    81 1    <int [5]>
3 Mathematics  82 1    <int [5]>
4 Music        62 1    <int [5]>
5 Physics      60 1    <int [5]>
6 Romance Studies 66 1    <int [5]>
7 Psychology & Neuroscience 59 1    <int [5]>
8 Art, Art History & Visual Studies 44 0.814 <int [5]>
9 Asian & Middle Eastern Studies 45 0.832 <int [5]>
10 Cinematic Arts 7 0.129 <int [5]>
```

11 Computer Science	48 0.888 <int [5]>
12 Dance Program	21 0.388 <int [5]>
13 English	50 0.925 <int [5]>
14 Gender, Sexuality & Feminist Studies	13 0.240 <int [5]>
15 History	47 0.869 <int [5]>
16 Philosophy	24 0.444 <int [5]>
17 Political Science	43 0.795 <int [5]>
18 Sociology	35 0.647 <int [5]>
19 Statistical Science	32 0.592 <int [5]>
20 Writing and Rhetoric Program	31 0.573 <int [5]>

Within each selected department i , we selected 5 faculty by simple random sampling without replacement. The probability that faculty member j was chosen, given that department i was selected, was therefore

$$\pi_{j|i} = \frac{5}{M_i}.$$

The overall inclusion probability for faculty member j in department i is

$$\pi_{ij} = \pi_i \pi_{j|i} = \pi_i \cdot \frac{5}{M_i}.$$

Data Collection

```

samples <- read.csv("pps_sample.csv")

samples <- samples |>
  select(-any_of(c("M_i", "pi_i", "pi", "weight"))) |>
  rename(
    woman = Sex...1...F..0...other.,
    grant = Ongoing.grant...1...Yes..0...No.,
    has_phd = PhD...1...Yes..0...No.,
    yrs_phd = Number.of.years.since.Phd,
    yrs_pub = Years.since.last.pub,
    intl_ugrad = International.undergrad.degree.
  )

# 3. attach M_i and pi_i, then build the weight
samples <- samples |>

```

```

left_join(dept_info, by = "department")

samples$weight <- (t_x / (n_dept * samples$M_i)) * (samples$M_i / 5)

# 4. survey design
fac_des <- svydesign(
  ids = ~department,
  weights = ~weight,
  data = samples
)

```

Let S be the set of sampled departments and s_i the set of sampled faculty in department i . For a variable y_{ij} , the estimated population total is

$$\hat{t}_y = \sum_{i \in S} \sum_{j \in s_i} w_{ij} y_{ij},$$

and the estimated population mean is

$$\hat{\bar{y}} = \frac{\sum_{i \in S} \sum_{j \in s_i} w_{ij} y_{ij}}{\sum_{i \in S} \sum_{j \in s_i} w_{ij}}.$$

The fraction of faculty who are women, the fraction with an ongoing grant, and the fraction with an international undergraduate degree are weighted proportions. The average years since Ph.D. and Our goal was a final sample of 100 faculty, which we obtained in two stages. In the first stage, we selected 20 of the 41 Trinity departments using systematic PPS sampling without replacement, with each department's number of faculty as its measure of size. In the second stage, we selected 5 faculty from each chosen department by simple random sampling without replacement. This gives a final sample of $20 \times 5 = 100$ faculty. the average years since last publication are weighted means, restricted to sampled faculty who hold a Ph.D. and to those who have published, respectively. We computed all estimates with the 'survey' package, using departments as the clusters, and report 95% confidence intervals.

Question 1. What is an estimate of the fraction of faculty who are women?

```

#|label: women-prop

prop_women <- svymean(~woman, fac_des)
prop_women

```

```

      mean      SE
woman 0.37 0.0567

```

```
confint(prop_women, level = .95)
```

```
      2.5 %    97.5 %  
woman 0.2588545 0.4811455
```

We estimate that 37.0% of Trinity faculty are women. We are 95% confident that the true fraction of Trinity faculty who are women is between 25.9% and 48.1%.

Question 2. What fraction of faculty have at least one grant that is ongoing in 2026?

```
#|label: grant  
  
prop_grant <- svymean(~grant, fac_des)  
prop_grant
```

```
      mean    SE  
grant 0.19 0.0552
```

```
confint(prop_grant, level = .95)
```

```
      2.5 %    97.5 %  
grant 0.08180416 0.2981958
```

We estimate that 19.0% of Trinity faculty have at least one grant ongoing in 2026. We are 95% confident that the true fraction of Trinity faculty with at least one grant ongoing in 2026 is between 8.2% and 29.8%.

Question 3. Among faculty who have a Ph.D. degree, what is the average number of years since they received the Ph.D. degree?

```
#|label: phd  
  
phd_des <- subset(fac_des, has_phd == 1)  
  
avg_phd <- svymean(~yrs_phd, phd_des)  
avg_phd
```

```
      mean    SE  
yrs_phd 25.703 2.5448
```

```
confint(avg_phd, level = .95)
```

```
      2.5 %    97.5 %  
yrs_phd 20.71508 30.69032
```

Among faculty who have a Ph.D., we estimate that the average time since receiving the degree is 25.7 years. We are 95% confident that, among Trinity faculty with a Ph.D., the true average number of years since receiving the degree is between 20.7 and 30.7 years.

Question 4.

a) Among those who have published papers, what is the avg number of years since their last publication?

```
#|label: pub-yrs  
  
pub_des <- subset(fac_des, !is.na(yrs_pub))  
  
avg_pub <- svymean(~yrs_pub, pub_des)  
avg_pub
```

```
      mean      SE  
yrs_pub 2.8481 0.6691
```

```
confint(avg_pub, level = .95)
```

```
      2.5 %    97.5 %  
yrs_pub 1.536777 4.159426
```

Among faculty who have published, we estimate that the average time since their last publication is 2.85 years. We are 95% confident that, among Trinity faculty who have published, the true average number of years since their last publication is between 1.54 and 4.16 years.

b) Did this faculty receive their undergraduate degree outside of the US?

```
#|label: int-ug  
  
prop_intl <- svymean(~intl_ugrad, fac_des, na.rm = TRUE)  
prop_intl
```

	mean	SE
intl_ugrad	0.18367	0.0614

```
confint(prop_intl, level = .95)
```

	2.5 %	97.5 %
intl_ugrad	0.06325132	0.3040956

We estimate that 18.5% of Trinity faculty with a known undergraduate institution received their undergraduate degree outside the US. We are 95% confident that the true fraction of these faculty who received their undergraduate degree outside the US is between 6.2% and 30.9%.

Caveats and other considerations

Systematic PPS sampling has a drawback for variance estimation. As with any systematic sample, some pairs of departments can never be selected together, so the joint inclusion probability π_{ik} is 0 for some pairs (i, k) . The usual Horvitz–Thompson variance estimator needs these joint probabilities, so it cannot be used to estimate the variance of our estimates. Following lecture, we instead pretend the sample was drawn with replacement and use the with-replacement variance estimator.

Initially, we considered a simple random sample (SRS) of 100 faculty. An SRS has equal inclusion probabilities and is easy to analyze. The problem is that it requires a complete list of every eligible Trinity faculty member before any sampling can happen, and the goal of our design was to avoid compiling faculty information from all 41 department websites, so an SRS did not fit our data collection.

We also considered stratified sampling, with each department as a stratum. Stratifying ensures every department is represented, and it can improve precision when faculty within a department are similar to each other but departments differ. However, department sizes vary a lot. With proportional allocation,

$$n_i = n \cdot \frac{M_i}{M_0},$$

many small departments would be allocated well under one faculty member. For example, Cinematic Arts, with 7 faculty, would get about 0.6. Rounding would either leave many small departments out or force us to move far from proportional allocation. Stratification would also still require a faculty list for all 41 departments.

We therefore chose a two-stage PPS cluster design, which balances statistical representation against practical effort. Sampling departments with probability proportional to size gives larger departments a better chance of being selected, and the second-stage SRS gives a probability

sample of faculty within each selected department. Most importantly, we only need to build faculty lists for the 20 selected departments instead of all 41. Choosing 20 departments, rather than fewer, also spreads the sample over many clusters, which limits the loss of precision that happens when faculty within the same department are similar.